# FLIGHT DELAY AND AIRPORT OPERATIONS

Presentation dialogue – Pipeline, validation and dashboard
Speaker handover begins after the Gold Layer explanation

**Estimated speaking time:** 10–15 minutes, depending on how much time you spend navigating the Job, notebook and dashboard.

## 1. Transition from the Gold Layer

“Thank you. As we have seen, the Gold Layer contains our dimension tables, fact tables and daily KPI mart, which are prepared for business reporting. I'll continue from here by explaining how we orchestrate the complete pipeline using Databricks Workflows, how we validate the pipeline outputs, and finally, how we present the results through our Flight Operations Command Center dashboard.”

## 2. Databricks Job – Pipeline Orchestration

“To automate the complete data processing flow, we created a Databricks Job named **capstone-03-flight_delay**. Instead of manually running each notebook one by one, we use this Job to execute the different stages in the correct order. Our workflow contains six tasks.”

### The six workflow tasks

• **bronze_adls_autoloader:** ingests flight schedule, flight movement and weather files from Azure Data Lake Storage into Bronze using Auto Loader.

• **bronze_azure_sql:** ingests reference data from Azure SQL, including aircraft, airline and airport master data.

• **data_profile:** runs after both ingestion tasks and profiles the ingested data.

• **silver_transform:** performs cleaning, validation, deduplication and data-quality checks.

• **gold_build:** creates Gold dimensions, fact tables and the KPI mart.

• **validate:** runs the validation notebook to check table availability, reconciliation and KPI consistency.

“The first two ingestion tasks are independent, so they can run in parallel. Once both are completed, data profiling runs, followed by Silver transformation, Gold creation and final validation. The overall flow is ADLS and Azure SQL ingestion, then profiling, Silver, Gold and validation.”

### DAG and task dependencies

“This workflow is represented as a DAG, which stands for Directed Acyclic Graph. In simple terms, it shows the tasks and their dependencies. We configured the dependencies so that each stage starts only after its required preceding tasks have completed successfully.”

### File Arrival trigger

“We also configured a File Arrival trigger for the Job. When a new file arrives in the configured Unity Catalog Volume location, the trigger can start the Databricks Job automatically. The File Arrival trigger starts the workflow, whereas Auto Loader discovers and ingests the files.”

“We use Auto Loader with the **availableNow** trigger. It processes the files currently available and then stops. So, our solution uses incremental, batch-oriented file ingestion rather than continuous real-time streaming.”

### Retry settings, notifications and monitoring

“We configured a two-minute retry delay and a maximum of three retries. This allows Databricks to automatically attempt a failed task again. We also configured notifications for Job start, successful completion and failure. The Job run history lets us review previous executions, see which tasks succeeded or failed, and investigate errors. The main benefit of Workflows is that it automates the pipeline, maintains execution order and makes the process easier to monitor and rerun.”

## 3. Pipeline Validation Notebook

“After the Gold Layer is built, we execute our final pipeline validation notebook. Its purpose is to verify that the complete pipeline has executed correctly and that the output data is consistent.”

“The notebook defines the catalog and schema locations for our Bronze, Silver and Gold tables. It uses helper functions: **count_rows()** counts records in a table, while **add_result()** records the outcome of each validation check as a pass or fail.”

### Bronze validation

“First, the notebook checks whether all six Bronze tables exist and contain records. This helps confirm that the source data has been successfully ingested.”

### Silver schedule reconciliation

“Next, we compare the number of records in Bronze flight schedules with the accepted and rejected records in Silver. The expected condition is: **Bronze schedule records = Accepted Silver schedule records + Rejected Silver schedule records.** This helps confirm that records have been accounted for during Silver processing and have not disappeared unexpectedly.”

### Gold table validation

“Next, the notebook checks whether all nine Gold tables exist and contain records. These include four dimension tables, four fact tables and the daily KPI mart. This confirms that the Gold Layer has been created.”

### KPI consistency validation

“We compare the flight and cancellation totals in **fact_flight_activity** with the corresponding totals in **mart_flight_daily**. The purpose is to make sure the aggregated KPI mart agrees with the underlying flight activity data.”

### Final validation report

“Finally, the notebook collects the validation results and displays a report showing how many checks passed and how many failed. The Gold notebook creates the business-ready tables, while this validation notebook checks whether the outputs are complete and consistent. This is important because we should not publish dashboard results without checking the underlying data.”

## 4. Flight Operations Command Center – Dashboard

“Now that the pipeline and validation are complete, I'll move to the reporting part of our project. We created an AI/BI dashboard in Databricks called **Flight Operations Command Center**. Its purpose is to help users understand flight operations through KPIs, charts and summary tables. The dashboard uses our Gold Layer tables as its data source.”

“We organized it into four pages: Executive Overview; Airport and Airline Performance; Fleet and Schedule; and Delays, Cancellations and Weather. We also have global filters for Date Range, Airport, Airline and Flight Status, which help users focus on the information they want to analyse.”

### Page 1 – Executive Overview

“The first page provides a high-level summary of flight operations. The KPI cards show Total Flights of 11,097, Cancelled Flights of 2,754, Departed Flights of 160, a Cancellation Rate of approximately 24.82%, and Average Departure Delay.”

“The **Daily Flight Operations Trend** shows how flight activity changes over time. **Flight Status Distribution** shows the distribution across statuses. **Top Origin Airports by Flight Volume** identifies airports with the highest number of

scheduled flights. The **Daily Cancellation Rate** chart shows how cancellation rates change across dates. This page is mainly designed for management users who want a quick overview."

### Page 2 – Airport and Airline Performance

"The second page focuses on airline and airport performance. **Airline Flight Volume vs Cancellations** compares flight volumes and cancellations across airlines. The **Airline Scorecard** provides airline name, total flights, cancelled flights and cancellation percentage."

"The **Airport Flight Activity** chart shows activity by origin airport. The **Movement Event Distribution** displays events such as departed, arrived, gate change and boarding started. Together, these visuals help users understand how activity and cancellations are distributed across airlines and airports."

### Page 3 – Fleet and Schedule

"The third page focuses on aircraft fleet information and scheduled flights. The KPI cards show 146 aircraft in the fleet, total seat capacity of approximately 24.47 thousand, approximately 11.1 thousand scheduled flights and an average route distance of approximately 934.95 kilometres."

"**Aircraft by Manufacturer** shows the distribution of manufacturers. **Fleet Status** provides an overview of aircraft status. **Scheduled Departures per Day** shows the scheduled departure pattern. **Scheduled Flights by Status** shows the distribution of scheduled flight statuses. We also have **Seat Capacity by Aircraft Type**, comparing Basic, Premium and Standard categories, and **Aircraft by Engine Type**. This page helps users understand fleet composition, aircraft capacity and scheduled operations."

### Page 4 – Delays, Cancellations and Weather

"The fourth page focuses on cancellations, delay reasons and weather observations. **Cancelled Flights per Day** shows the daily cancellation trend. **Cancellation Rate by Origin Airport** helps identify airports with higher cancellation percentages. **Delay Reasons by Airport** displays recorded reasons such as ATC, Technical, Weather and None."

"The **Weather Condition Distribution** shows conditions including Wind, Clear, Rain and Fog. **Weather Observations by Airport** provides observation count, average visibility, average wind speed and precipitation. These visuals allow users to explore cancellations, recorded delay reasons and weather observations together. However, observing weather and cancellation patterns together does not prove that weather caused a cancellation; it only identifies patterns for further investigation."

## 5. Important Data Observations and Limitations

"While presenting the dashboard, I would also like to highlight a few important observations. Although we have 11,097 flight records, only 160 records have a recorded actual departure timestamp. The delay data also contains unusual values: 159 records have exactly 15 minutes of delay, and one record has a negative delay of -778 minutes."

"Because of these data-quality issues, the average delay and punctuality figures should be interpreted carefully. They need further investigation before being treated as reliable operational performance measures. Our weather observations cover only July 1 and July 2, 2026, so they represent a limited observation period. A dashboard should not only display numbers; we should also understand the quality and coverage of the data behind them."

## 6. Dashboard Publishing and Final Handover

"Once the dashboard is ready, we can publish it so intended users can view it and share it with relevant stakeholders with viewing access. The dashboard provides a business-friendly view of the data, while the underlying Gold tables remain available for deeper analysis through Databricks SQL."

"We have also prepared a Genie space with SQL-grounded questions so users can explore flight operations data using natural-language questions. Its response validation is still pending, so we treat it as an additional exploration capability rather than a fully validated production feature."

"To summarize my part: we use Databricks Workflows to orchestrate the complete pipeline, from ingestion through Silver and Gold processing. We use the validation notebook to check table availability, record reconciliation and KPI consistency. Finally, we use the Flight Operations Command Center dashboard to present the processed data through KPIs, charts and operational summaries. Together, these components make our solution an end-to-end data analytics

pipeline, from source ingestion to business reporting. Thank you."

# 7. Technical Challenges – SME Discussion

Use these examples if the SME asks what technical challenges you faced. Keep the wording factual and do not claim a specific permission grant unless you have verified it.

## A. File Arrival trigger and Azure Storage permissions

"One technical challenge was configuring the File Arrival trigger. We had a storage access permission issue when Databricks needed to monitor the configured storage location. The Azure role commonly involved in this setup is **Storage Blob Data Contributor**, assigned to the Databricks Access Connector managed identity at the appropriate storage scope. We investigated the storage access configuration so the trigger could monitor the location."

**Accuracy note:** Mention Storage Blob Data Contributor as the role you granted only if you have verified it in Azure IAM role assignments. The project notebooks do not confirm the exact role assignment. This Azure role is distinct from Unity Catalog privileges such as USE CATALOG, USE SCHEMA and READ VOLUME.

## B. Conflicting schedule records

"We found 37 flight IDs with conflicting schedule records, covering 74 rows. Because there was no reliable effective-time field to determine which version was correct, we quarantined all 74 records instead of selecting one arbitrarily."

## C. Duplicate handling

"We had to identify duplicates without deleting valid records. We applied different rules depending on the dataset: exact business-record matching for schedules and business-key-based deduplication for the other datasets."

## D. Data relationships

"We checked whether operational records referenced valid airline, aircraft and airport master records. Referential-integrity checks helped us identify orphan records for review rather than silently removing them."

## E. Workflow dependencies

"We needed to arrange the notebooks in the correct order. We configured task dependencies so the two ingestion tasks could run independently, followed by profiling, Silver transformation, Gold creation and final validation."

## F. Dashboard KPI reliability

"The main dashboard challenge was making sure the KPIs represented the data correctly. Only 160 records had actual departure timestamps, and the delay data included unusual values. We therefore highlighted these limitations rather than presenting the delay and punctuality metrics as fully reliable."

## 8. Likely SME Questions – Short Answers

| SME question | Short answer |
|---|---|
| Why did you quarantine all 74 schedule records? | Because 37 flight IDs had conflicting records and we had no reliable effective-time field to select the correct version. |
| How did you prevent data loss? | Through Bronze-to-Silver reconciliation and Gold KPI consistency checks. |
| Why use Databricks Workflows? | To automate notebook execution, manage dependencies and monitor pipeline runs. |
| Is the pipeline real-time? | No. It uses incremental, batch-oriented ingestion with Auto Loader's availableNow trigger. |
| Difference between File Arrival trigger and Auto Loader? | The File Arrival trigger starts the Job when files arrive; Auto Loader discovers and ingests the files. |
| What is a DAG? | A Directed Acyclic Graph that represents workflow tasks and their dependencies. |
| Are all dashboard KPIs reliable? | Not all. Delay and punctuality metrics need investigation because actual departure data is limited and contains anomalies. |
| Does weather cause cancellations? | The dashboard can show patterns, but correlation does not establish causation. |
| What does Storage Blob Data Contributor do? | It grants the assigned identity data-plane permissions to read, write and manage blob data within the assigned storage scope. Say it was the exact role granted only if verified. |

## 9. Final Presentation Reminders

• Start from the Gold Layer handover; do not repeat the full project introduction already covered by your colleague.

• On the Job screen, point out the six tasks and explain that the two ingestion tasks can run in parallel.

• On the validation notebook, emphasize Bronze-to-Silver reconciliation and Gold KPI consistency.

• On the dashboard, explain the business purpose of each page rather than reading every chart title mechanically.

• Be transparent about limited actual departure timestamps, unusual delay values and the two-day weather coverage.

• Do not claim Genie response validation is complete.

• Do not claim an exact Azure role assignment unless you have checked the Access Connector's IAM role assignments.